Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	E-Commerce Order Management Database System
Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to describe the requirements of the E-Commerce Order Management Database System. The system is developed to manage online shopping activities, including customer registration, product management, order processing, payment handling, and delivery tracking in an organized and efficient manner.

1.2 Project Scope

The database system stores and manages information related to customers, products, orders, payments, and deliveries. It helps automate the complete order process from product selection to successful delivery while maintaining accurate records.

1.3 Intended Users

- Customers – Create accounts, browse products, place orders, make payments, and check order status.
- Administrators – Manage products, customers, orders, and generate reports.
- Inventory Managers – Monitor stock availability and update inventory.
- Delivery Executives – Update shipment progress and delivery confirmation

2. Business Requirements

The following are the key business needs the system must satisfy:

- Maintain complete customer records.
- Store product details, prices, categories, and stock information.
- Allow customers to order multiple products in a single order.

- Calculate the total bill automatically.
- Record payment details and transaction status.
- Monitor order progress from order placement to delivery.
- Reduce product stock after order confirmation.
- Generate reports for sales, customers, and inventory.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow new customers to register with personal information.
FR-2	The system shall allow administrators to manage product details.
FR-3	The system shall allow customers to search and view products.
FR-4	The system shall allow customers to place orders for multiple items.
FR-5	The system shall automatically calculate the total order value.
FR-6	The system shall record payment method, transaction ID, and payment status.
FR-7	The system shall reduce product stock after successful order confirmation.
FR-8	The system shall update order and delivery status.

4. Non-Functional Requirements

- Performance: The system should process customer requests within 2–3 seconds.
- Security: Customer data and payment details must be protected using secure encryption.
- Reliability: The database should ensure consistency and avoid duplicate or invalid records.
- Usability: The application should provide an easy-to-use interface.
- Scalability: The system should handle future growth in products, customers, and orders.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	**Data Type**	**Description**
customer_id	INT (Primary Key)	Unique product identifier
full_name	VARCHAR(100)	Name of the product
email	VARCHAR(100)	Unique email used for login/contact
phone_number	VARCHAR(15)	Customer contact number
address	VARCHAR(255)	Shipping address of the customer
registration_date	DATE	Date the customer account was created

6. Assumptions and Constraints

- The database will be implemented using MySQL, PostgreSQL, or Oracle.
- Customers must register before placing orders.
- Each order is associated with only one customer.
- Orders cannot be placed for unavailable products.
- Every payment record corresponds to one order.
- Only administrators can update product information.

7. Conclusion

This Requirement Specification Document outlines the functional, business, and non-functional requirements of the E-Commerce Order Management Database System. It serves as the basis for designing the database schema, ER diagram, tables, relationships, and implementation of the project.